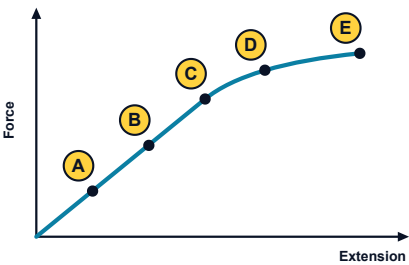


Forces and elasticity

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

QUESTION	ANSWER	✓
1 Give three ways in which the shape of an object can be changed.	Stretching, compressing or bending	☐
2 What is elastic deformation?	The object returns to its original shape when the forces are removed.	☐
3 What is inelastic deformation?	The object does not return to its original shape when the forces are removed.	☐
4 State how the extension of a spring depends on the force applied, below the limit of proportionality.	The extension is directly proportional to the force.	☐
5 Describe the relationship between the force on an elastic object and its extension.	As the force increases, the extension increases.	☐
6 What is the limit of proportionality of an elastic object?	The point beyond which extension is no longer directly proportional to force.	☐
7 Which letter shows the limit of proportionality on this force–extension graph? 	C	☐
8 State the standard unit of force.	Newton (N)	☐
9 State the standard unit of spring constant.	Newton per metre (N/m)	☐
10 Define spring constant.	The force needed to stretch (or compress) a spring by 1 m.	☐

QUESTION		ANSWER	✓
11	State the standard unit of extension.	Metre (m)	☐
12	State how the extension of an elastic object is calculated.	Extension = stretched length – original length	☐
13	Name the energy store that increases when an elastic object is stretched or compressed.	The elastic potential energy store	☐
14	Describe how to find the spring constant from a graph of force (y-axis) against extension (x-axis).	It is the gradient of the straight section.	☐
15	Describe how to find the spring constant from a graph of extension (y-axis) against force (x-axis).	It is $1 \div$ gradient of the straight section.	☐
16	State the standard unit of elastic potential energy.	Joule (J)	☐
17	State the equation linking force, spring constant and extension.	force = spring constant $\times$ extension ($F = k e$)	☐
18	Explain why more than one force is needed to stretch or compress a stationary object.	A single force would just make the object move.	☐
19	State how the work done stretching a spring compares with the elastic potential energy stored.	They are equal (if the spring is not inelastically deformed).	☐
20	State how linear and non-linear relationships look on a force–extension graph.	Linear: a straight line. Non-linear: a curve.	☐